

Fixed-Ro equal-area theory on the equatorial β plane, for the SS09 shallow-water model

pengcheng-ss09 suite, Tier 0 (A0 memo)

2026-07-16

Beta-plane translations of the closed forms in the fixed-Ro manuscript (`papers/fixed-ro-eq-area_ms-draft-v1.pdf`, “ms” below), derived against this model’s exact equations. Every result carries a basis tag; “pending run” tags get upgraded as the suite’s runs confirm them. Section 8 lists the falsifying checks and their status, and the run record follows it.

1 Setup and notation

Steady state, hemispherically symmetric forcing ($y_0 = 0$), northern hemisphere ($y > 0$). The model’s steady equations (`SCIENCE.md` section 2), with $T = \theta/1.6$ and the v equation’s factor of two multiplying only the time tendency (so it drops out of every steady balance):

$$v \left(\beta y - \frac{\partial u}{\partial y} \right) = \mathcal{H}(\theta_E - \theta) u \frac{\partial v}{\partial y} + \epsilon_u u + v_d \mathcal{H}(u) \operatorname{sgn}(y) \frac{\partial u}{\partial y}, \quad (1)$$

$$\beta y u = -\frac{gH}{T_0} \frac{\partial T}{\partial y} + k_v \frac{\partial^2 v}{\partial y^2}, \quad (2)$$

$$\frac{\delta \Delta_z}{H} \frac{\partial v}{\partial y} = \frac{\theta_E - \theta}{\tau}. \quad (3)$$

Forcing (SS09 parabolic profile, the theory-matched choice): $\theta_E = \theta_{00} - \Delta_y (y/y_1)^2$ for $|y| < y_1$. In temperature units,

$$T_E(y) = T_E(0) - b y^2, \quad b \equiv \frac{\Delta_T}{y_1^2}, \quad \Delta_T \equiv \frac{\Delta_y}{1.6}, \quad (4)$$

with $T_0 = 300$ K. The small-angle limit of the ms is exact here: the β plane has no $\cos \varphi$ factors, so no truncation enters any of the following.

2 The one-line mapping

Every ms result carries over with the substitutions

$$\operatorname{Ro}_{\text{th}} \rightarrow R_\beta \equiv \frac{4gH\Delta_T}{T_0\beta^2 y_1^4}, \quad \Delta_h \theta_0 \rightarrow \Delta_T, \quad a\varphi \rightarrow y, \quad \varphi_{\text{Ro}} \rightarrow Y_{\text{Ro}}/y_1. \quad (5)$$

[**derived**] R_β follows from the sphere definition $\operatorname{Ro}_{\text{th}} = gH\Delta_h/(\Omega a)^2$ with $\Omega = \beta a/2$ and $\Delta_h = \Delta_T a^2/(T_0 y_1^2)$ (matching the forcing curvature at the equator); the planetary radius cancels, as it must on a β plane. At this repo’s defaults ($g = 9.81 \text{ m s}^{-2}$, $H = 16 \text{ km}$, $\Delta_y = 50 \text{ K}$ so $\Delta_T = 31.25 \text{ K}$, $T_0 = 300 \text{ K}$, $\beta = 2 \times 10^{-11} \text{ m}^{-1} \text{ s}^{-1}$, $y_1 = 9439 \text{ km}$):

$R_\beta = 0.0206.$

(6)

3 Wind and temperature

[derived: pending run confirmation] Uniform $\text{Ro} = (\partial_y u)/(\beta y)$ integrates to $u_{\text{Ro}}(y) = \text{Ro} \beta y^2/2$ (ascent at $y = 0$). Gradient balance (neglecting k_v) gives $\partial_y T = -(T_0/gH) \text{Ro} \beta^2 y^3/2$, so

$$T_{\text{Ro}}(y) = T_{\text{Ro}}(0) - c y^4, \quad c \equiv \frac{\text{Ro} T_0 \beta^2}{8gH}. \quad (7)$$

4 Equal-area edge and equatorial temperature

[derived: confirmed at the 1–8% level by runs 1a–1b below] With $D(y) = T_{\text{Ro}}(y) - T_E(y)$, imposing $D(Y) = 0$ and $\int_0^Y D dy = 0$ (measure dy ; the Newtonian relaxation has uniform τ) gives $\frac{2}{3}bY^2 = \frac{4}{5}cY^4$, i.e. $Y^2 = \frac{5}{6}b/c$, which in Section-2 variables is

$$Y_{\text{Ro}} = y_1 \left(\frac{5R_\beta}{3\text{Ro}} \right)^{1/2} \quad [\text{ms Eq. 13 analog}], \quad (8)$$

$$T_E(0) - T_{\text{Ro}}(0) = \frac{5}{18} \frac{R_\beta \Delta_T}{\text{Ro}} \quad [\text{ms Eq. 15 analog}]. \quad (9)$$

Defaults, $\text{Ro} = 1$: $Y_{\text{Ro}} = 1.75 \times 10^6$ m and equatorial depression 0.18 K (in T ; multiply by 1.6 for θ units). At $\text{Ro} = 0.15$ the depression is 1.2 K (T) / 1.9 K (θ).

Consistency note: the analogous derivation for the $\sin^2$ profile needs the numerical equal-area solver (puffins `equal_area_bouss` with Ro-scaled `del_h_over_ro`, verified in puffins PR #46). Near the equator $\sin^2(\pi y/(2y_1)) = (\pi y/(2y_1))^2 + O(y^4)$, so $\sin^2$ behaves like a parabola with curvature enhanced by $(\pi/2)^2 = 2.47$; Section 7's postdiction check uses exactly this factor.

5 Meridional wind, heat flux, momentum flux

[derived: v magnitude and shape confirmed at the 1–4% level by runs 1a–1b below]

The steady thermodynamic balance (3) integrates to the model's v profile directly, using $\theta_E - \theta = 1.6(T_E - T) = -1.6D$:

$$v(y) = -\frac{1.6H}{\delta \Delta_z \tau} \int_0^y D dy' = \frac{1.6H}{\delta \Delta_z \tau} \frac{5}{18} \frac{R_\beta \Delta_T}{\text{Ro}} Y_{\text{Ro}} (x - 2x^3 + x^5), \quad x \equiv \frac{y}{Y_{\text{Ro}}}. \quad (10)$$

The bracket is exactly the ms Eq. (16) shape; it peaks at $x = 5^{-1/2} = 0.447$ with value 0.286, and $v_{\text{max}} \propto \text{Ro}^{-3/2}$ (ms Eq. 20 analog). Defaults, $\text{Ro} = 1$: $v_{\text{max}} = 3.0 \times 10^{-3} \text{ m s}^{-1}$ at $y = 0.78 \times 10^6$ m.

Column fluxes, using the two-layer reading (upper slab of depth δ with velocity $+v$ and potential temperature θ_{top} , return slab of depth δ with $-v$ and θ_{bot} , $\theta_{\text{top}} - \theta_{\text{bot}} = \Delta_z$; return-flow u negligible):

$$\text{heat flux: } \int v \theta dz = \delta \Delta_z v(y) \quad [\text{ms Eq. 16}], \quad (11)$$

$$\text{momentum flux: } \int v u dz = \delta v(y) u_{\text{Ro}}(y) \quad [\text{ms Eq. 17}], \quad (12)$$

$$\text{vertical velocity: } w = \delta \frac{\partial v}{\partial y}, \quad w_{\text{max}} \propto \text{Ro}^{-1} \quad [\text{ms sec. 3f}]. \quad (13)$$

The ms's counterintuitive momentum-flux result carries over: expanding $\delta v u_{\text{Ro}}$, the leading term in y is Ro-independent (the Ro^{-1} strengthening of v cancels the Ro weakening of u in the product) and $\partial(\text{flux})/\partial \text{Ro} < 0$ strictly inside the cell, exactly as verified for the sphere forms in puffins `test_eq_area.py`.

6 Diagnosed quantities: conventions for the suite

- **Local Ro(y):** $(\partial_y u)/(\beta y)$, the model’s existing diagnostic (NaN within $|y| < \Delta y_{\text{grid}}$ of the equator).
- **Cell-mean Ro, two definitions, both reported** (paralleling Hill et al. 2025): (i) the area mean of local Ro over $[y_{\text{Ro-max}}, Y_{\text{edge}}]$; (ii) the best-fit Ro regressing $u(y)$ against $\beta y^2/2$ over the same interval. The near-ascent region is excluded because $\text{Ro} \rightarrow 0$ at the equator whenever $v_d > 0$ (Zhang et al. 2025, $u \sim y^3$).
- **Cell edge, three definitions, all reported:** the v -threshold edge ($|v|$ below 10% of its extremum, existing diagnostic; see run 1a for its failure mode at small v_d), the jet latitude (existing spline diagnostic), and the θ -merge latitude (where θ rejoins θ_E within a tolerance, searched poleward of the *jet*; see run 1a). The theory says they coincide; their splitting is itself a result.
- **Momentum-budget shares:** each steady term of (1) evaluated from output; the drag share $\epsilon_u u$ and the vertical-advection share quantify the accuracy of the EMFD–Coriolis balance underlying Zhang et al. (2025) Eq. 24, $\text{Ro} = v/(v + v_d)$.

7 Emergent cell-mean Ro closure

[**derived: pending run confirmation**] Combining Zhang’s Eq. 24 at the cell-mean level with Section 5’s v magnitude (the cell mean of the bracket over $[0, 1]$ is $1/6$):

$$\bar{\text{Ro}} = \frac{\bar{v}}{\bar{v} + v_d}, \quad \bar{v}(\bar{\text{Ro}}) = \frac{1.6H}{\delta\Delta_z\tau} \frac{5}{18} \frac{R_\beta\Delta_T}{\bar{\text{Ro}}} \frac{Y_{\text{Ro}}(\bar{\text{Ro}})}{6}, \quad (14)$$

a transcendental equation for $\bar{\text{Ro}}(v_d)$ with no simulation input. This is the suite’s analysis G; it extends Zhang et al. (2025) section 4 (their max-Ro scalings $v_d^{-2/5}$, $v_d^{-1/4}$) to the cell-mean variable the ms uses.

Postdiction check [**computed**]: Section 5’s v with the $\sin^2$ profile’s larger near-equator curvature ($(\pi/2)^2 = 2.47$ in b , entering v through $b^2 Y/c$ as a factor ~ 9.6) predicts $v_{\text{max}} \sim 0.03 \text{ m s}^{-1}$ for the published defaults, matching Zhang et al. (2025) Figs. 2–3 magnitudes within $\sim 1.5\times$.

8 Falsifying checks and status

1. **Edge, equatorial depression, v profile vs diagnosed Ro (the most-falsifying single check):** SS09-parabolic forcing, $v_d = 0$. Framing correction from the first run: with the default vertical momentum advection on, $v_d = 0$ is *not* the $\text{Ro} = 1$ limit (Zhang et al. 2025: $u \sim \beta y^2/3$ near the ascent, and drag lowers it further); the theory must be evaluated at the run’s diagnosed cell-mean Ro. The $\text{Ro} \rightarrow 1$ anchor requires **--no-vert-advect-u** and reduced drag (run 1b). Status: runs 1a–1b below.
2. **Edge collapse:** $Y_{\text{edge}}(\text{Ro}/R_\beta)^{1/2}/y_1 = (5/3)^{1/2}$ flat across the v_d , Δ_y , β , and radius ladders. Status: pending Tiers 1–2.
3. **Radius-ladder null:** dimensional Y_{edge} invariant under $(\beta \rightarrow \beta/k, y_1 \rightarrow ky_1, L \rightarrow kL)$. Status: pending Tier 2.

4. **v -profile shape and $\text{Ro}^{-3/2}$ magnitude scaling** (Section 5). Status: pending Tier 1.
5. **Momentum flux increases as Ro decreases** (Section 5). Status: pending Tier 1.
6. **Emergent $\bar{\text{Ro}}(v_d)$** (Section 7). Status: pending Tier 1 + analysis G.

Run record

Each confirming or refuting run gets an entry: configuration, output path, measured vs predicted.

2026-07-16, check 1a: $v_d = 0$, default physics (vertical advection on)

Config: SS09 parabolic forcing, $v_d = 0$, $n_y = 801$, $\Delta t = 30$ s, numba backend, steady-state stop (window 30 d, threshold 10^{-4}), converged day 959. Output: `model_output/fixed_ro_suite/tier0_amc_edge/amc_ss09_ny801_dt30_vd0.nc`. Steadiness caveat [**computed**]: the jet was still drifting +0.4% per 60 d at stop (drag-timescale tail), so numbers carry roughly a 2% allowance.

Diagnosed: cell-mean Ro 0.469 (area mean) / 0.470 (u regression); local $\text{Ro}(y)$ is *not* uniform: ~ 0.68 at the ascent (the vertical-advection 2/3 of Zhang et al. 2025, drag-reduced), declining through ~ 0.6 mid-cell to ~ 0.2 approaching the jet, i.e. the ms’s linear- Ro motivation in miniature.

Theory at diagnosed $\text{Ro} = 0.470$ vs run [**computed**]:

quantity	predicted	measured	rel. gap
equatorial depression $T_E(0) - T(0)$	0.381 K	0.379 K	0.5%
$v_{\max}$	9.28×10^{-3} m/s	9.39×10^{-3} m/s	1.2%
y of $v_{\max}$ ($0.447 Y_{\text{Ro}}$)	1.14×10^6 m	1.18×10^6 m	3.5%
cell edge Y_{Ro} (vs jet latitude)	2.55×10^6 m	2.36×10^6 m	−7.4%

Sections 4–5 upgraded to [**confirmed at the 1–8% level, single run**] for these observables at this Ro . Additional findings: the u plateau outside the cell is the parabolic forcing’s RCE gradient wind,

$$u_{\text{RCE}} = \frac{2gH\Delta_T}{T_0\beta y_1^2} = 18.35 \text{ m/s}, \quad (15)$$

constant out to y_1 (verified to 3 digits); the v -threshold edge diagnostic is unusable at small v_d because the drag circulation $v = \epsilon_u u_{\text{RCE}}/(\beta y) \sim 2 \times 10^{-3}$ m/s exceeds 10% of $v_{\max}$ out to $\sim 9.3 \times 10^6$ m; the θ -merge diagnostic must search poleward of the jet, not of the v max, to avoid the equal-area lobe crossing.

2026-07-16, check 1b: $v_d = 0$, vertical advection off

Config: as 1a plus `--no-vert-advec-u`; converged day 971. Output: `.../tier0_amc_edge/amc_ss09_ny801_dt30_vd0_novert.nc`.

Removing vertical advection left the cell-mean Ro essentially unchanged (fit 0.473 vs 0.470; area mean 0.536 vs 0.469), because **the Rayleigh drag, at its default $\epsilon_u = 10^{-8} \text{ s}^{-1}$, is the leading-order non-AMC agent for this forcing** [**computed**]: the steady zonal momentum balance over the cell interior is

$$v \left(\beta y - \frac{\partial u}{\partial y} \right) = \epsilon_u u \quad \text{to within } \sim 5\text{--}10\% \quad (16)$$

(median ratio 1.05 over $0.3 \times 10^6 < y < 2.2 \times 10^6$ m). The parabolic forcing's weak circulation ($v_{\max} \sim 9 \times 10^{-3}$ m/s) makes $(1 - \text{Ro}) \sim \epsilon_u u / (v \beta y)$ order one. Local $\text{Ro}(y)$ declines monotonically from ~ 1 at the ascent to ~ 0.2 at the jet.

Theory at diagnosed $\text{Ro} = 0.473$ vs run **[computed]**: depression 0.378 predicted vs 0.358 K measured (5.3%); $v_{\max}$ 9.18×10^{-3} vs 8.81×10^{-3} m/s (4.2%); edge 2.54×10^6 vs jet 2.36×10^6 m (−7.1%).

Consequences for the suite: (i) the $\text{Ro} \rightarrow 1$ anchor needs reduced ϵ_u , so the Tier-3 drag ladder is load-bearing rather than optional; (ii) at the $\sin^2$ forcing the circulation is $\sim 3\times$ stronger, so the drag share is $\sim 3\times$ smaller but still non-negligible at $v_d = 0$ (consistent with Zhang et al. 2025 Fig. 4's max $\text{Ro} \sim 0.9$ there); (iii) the steady-state detector (window 30 d, 10^{-4}) fires while the jet still drifts $\sim 0.4\%$ per 60 d on the drag timescale, so budget-grade runs need a stricter threshold or a fixed post-detection extension.